

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 7: Polar Stereographic Projection (Polar Navigation)

Syllabus Source: CAE Oxford (pp. 249-286) | Pass Target: $\geq 90\%$ AviationExam

1. THE POLAR STEREOGRAPHIC ARCHITECTURE

The Polar Stereographic is an azimuthal conformal projection where the projection plane is tangent at the geographic pole, with the perspective projection point at the opposite pole:

- Meridians & Parallels: Meridians are straight lines radiating outward from the pole at their true angular spacing (360°). Parallels are concentric circles centered on the pole.
- Constant of the Cone: The convergence factor $n = 1.0$! Chart Convergence = $d\text{long} \times 1.0 = d\text{long}$. Chart convergence equals the full change of longitude everywhere on the chart!
- Great Circles on Polar Stereographic: Great Circles are virtually STRAIGHT LINES near the pole (technically circular arcs concave to the pole).
- Scale Factor (SF): $SF = 1.0$ at the pole. Scale expands away from the pole: Scale Factor = $2 / (1 + \sin(\text{lat}))$. At 60°N , $SF = 2 / (1 + 0.866) = 1.07$ (+7% expansion).

2. TRANSPOLAR FLIGHT OPERATIONS

Polar Stereographic charts are mandatory for transpolar airline routes (e.g. Europe to Far East over the Arctic Ocean), where standard magnetic compasses and Mercator charts become completely unusable.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Polar Stereographic Track & Con

Key Geometric Properties: - Planar azimuthal projection tangent to the pole, projected from opposite pole. - Cone constant $n = 1.0$! Chart convergence equals the entire change of longitude (Chlong). - Meridians are straight radial lines radiating from the pole. - Straight line tracks across the pole change true heading by exactly the difference in longitude between entry and exit!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Track Change on Polar Stereographic Chart

SCENARIO (Transpolar Flight Planning Drill): An aircraft is flying a straight track across a North Polar Stereographic chart: - Crossing Meridian 030°W , the measured true track is 080° True QUESTION: What will be the true track of the aircraft when it crosses Meridian 050°E along the same straight line? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Change in Longitude (dlong): - Longitude 1 = 030°W - Longitude 2 = 050°E - Points are in opposite hemispheres, so ADD: $d\text{long} = 30^\circ + 50^\circ = 80^\circ$. Step 2: Calculate Chart Convergence on Polar Stereographic ($n = 1.0$): - Chart Convergence = $d\text{long} \times n = 80^\circ \times 1.0 = 80.0^\circ$! Step 3: Calculate the New Track crossing 050°E : - Flying from West longitude to East longitude across the pole, the track angle decreases: - Track at $050^\circ\text{E} = \text{Initial Track} - \text{Convergence}$ - Track = $080^\circ - 80^\circ = 000^\circ$ True (due North)! FINAL ANSWER: True track at 050°E is 000° True.

[!] EXAM TRAP (AVIATIONEXAM) - The Polar Stereographic n = 1.0 Trap

- On a Polar Stereographic chart, n is ALWAYS 1.0! - Chart Convergence ALWAYS equals Change of Longitude (dlong)! - This is the simplest chart calculation in ATPL navigation.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a Polar Stereographic chart, what is the value of the Convergence Factor (Constant of the Cone n)?**

- [A]: 0.50
- [B]: EXACTLY 1.0
- [C]: 0.707
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The plane is tangent at the pole (latitude 90°); $\sin(90^\circ) = 1.0$. Therefore $n = 1.0$, and chart convergence equals dlong.

Q2: How do parallels of latitude appear on a Polar Stereographic projection?

- [A]: Straight parallel lines
- [B]: CONCENTRIC CIRCLES centered on the pole
- [C]: Curved lines concave to equator
- [D]: Hyperbolas

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Parallels form concentric circles around the pole, with spacing increasing slightly away from the center.

Q3: Where is the Scale Factor (SF) equal to exactly 1.0 on a direct Polar Stereographic chart?

- [A]: At the Equator
- [B]: AT THE POLE (Point of Tangency)
- [C]: At 60° latitude
- [D]: At the opposite pole

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The projection plane is tangent to the Earth at the pole, where there is zero scale distortion ($SF = 1.0$).